TLV Cash Register Interface Specification - Transparent Mode

Product name	TLV Cash Register Interface - Transparent Mode
Product number	-
Title	TLV Cash Register Interface Specification - Transparent Mode
Author	Robert Demming
Version	v1.0.2
Status	Release
Classification	Confidential
Info	Saved as "*TLV Cash Register Interface Specification - Transparent Mode.docx"*, on 2017-10-12 15:11

Copyright Statement © 2016-2017 Chess pT International B.V

All rights reserved. Disclosure to third parties of this document or any part thereof, or the use of any information contained therein for purposes other than provided for by this document, is not permitted, except with the prior and express written permission of Chess payment Technology International B.V.

Document history

Date	Author	Status	Remarks
2017-09-05	rdem	Preview	First version for review
2017-09-15	rdem		v1.0.1: Some tag number changed because tags numbered XX00 should not be used.
2017-10-06	rdem		v1.0.2: Fixed typos. Added comment that the card authorisation key is encrypted.

References

[1]	TLV Cash Register Interface Specification	v1.0.6
[2]	AN10833 MIFARE Type Identification Procedure (NXP) http://www.nxp.com/docs/en/application-note/AN10833.pdf	V3.6
[3]	http://nfc-tools.org/index.php?title=ISO14443A	

Table of Contents

1 Overview 4
1.1 Introduction 4
1.2 Prerequisites 4
2 Message Definitions 5
2.1 Transaction Request and Transaction Response 5
2.1.1 Transparent Mode Group 6
2.1.2 Transaction Result 6
2.2 Transparent Mode Request and Response 7
2.2.1 Transparent Mode 7
2.2.2 Card Detection Time-out 7
2.2.3 Card UID 7
2.2.4 Card ATR 7
2.2.5 Card Type 8
2.2.6 UI Control (Group) 8
2.2.7 Result Code 9
2.3 UI Control Request and Response 9
2.3.1 Display Text 9
2.3.2 Beep Type 9
2.3.3 LED Control 9
2.3.4 Result Code 9
2.4 Authenticate Request and Response 10
2.4.1 Key 10
2.4.2 Result Code 10
2.5 Read Card Data Request and Response 10
2.5.1 Address 11
2.5.2 Data Size 11
2.5.3 Data 11
2.5.4 Result Code 11
A Enumerations 12
A.1 Message Types 12
A.2 Result Codes 12

1 Overview

1.1 Introduction

The Chess pT cash register interface is a software interface that makes it possible to connect electronic cash registers (ECR) and vending machines with an Ingenico payment terminal provided by Chess pT.

The “Transparent Mode” of the cash register interface makes it possible to directly interact with custom NFC cards (non-EMV cards) by sending commands to the card after which the result is send back to the ECR. Further it is possible to control the UI elements of the terminal (display, LEDs and buzzer) while in transparent mode.

Transparent mode currently supports the following card types:

- MIFARE Ultralight (inc. EV1 and C)

In future versions support is planned for the following card types:

- MIFARE Mini
- MIFARE Classic 1k
- MIFARE Classis 4k
- MIFARE DESFire (incl. EV1, EV2)

This document only describes the “Transparent Mode” extension to the “TLV Cash Register Interface”. The base TLV interface is described in “TLV Cash Register Interface Specification” document [1].

1.2 Prerequisites

This manual is intended for software developers and it is assumed that the base “TLV Cash Register Interface Specification” document [1] is known.
Developers should have access to NXP document [2] for identification of the card type.

2 Message Definitions

This chapter describes the “Transparent Mode” messages in detail. The generic parts of a message, the header and footer, are described in the “TLV Cash Register Interface Specification” document [1]. In this chapter only the body parts of the different messages are described.

2.1 Transaction Request and Transaction Response

The *TransactionRequest* message is sent when the cash register wants the terminal to perform a transaction. This is an existing ECR command and has been extended with fields to enable transparent mode. In case an EMV-card (bank card) was presented a regular C-TAP transaction is performed and the terminal sends a *TransactionResponse* message to the cash register with the transaction outcome and the transaction receipt(s).
When a custom card was presented, a “Transparent Mode” group is returned instead and the terminal is in transparent mode after which TM-commands can be sent to the terminal. To leave transparent mode, send a *TransparentModeRequest* message indicating to stop transparent mode.

Table 1 describes the structure of the *TransactionRequest* and *TransactionResponse* messages. The *italic* rows are existing fields. The non-italic rows are the new “transparent mode” fields.

			Transaction	
Data Element	**Tag**	**Format**	**C-**	**R-**
Message Body	E2	GRP	M	M
• *Transaction ID*	*1A*	*16x*	*M*	*E*
• *Transaction Data*	*A3*	*GRP*	*M*	
○ *Amount*	*50*	*8i*	*M*	
○ *Currency Code*	*51*	*4i*	*O*	
○ *Currency Exponent*	*52*	*2i*	*O*	
○ *Progress Report Language*	*53*	*2a*	*O*	
○ *Transaction Type*	*54*	*2x*	*O*	
○ *Transaction Reference*	*55*	*0..16a*	*O*	
○ *Use Intermediate Result*	*58*	*2x*	*O*	
• *Print Data*	*A5*	*GRP*	*M*	
○ *Print Origin*	*4E*	*2x*	*M*	
○ *Paper Width*	*4F*	*2i*	*O*	
• *Terminal State*	*19*	*16b*		*M*
• *Transaction Result*	*56*	*2x*		*M*
• *Ticket Data*	*A6*	*TBL*		*O*
○ *Table Record*	*EF*	*REC 0..3*		*M*
▪ *Ticket Type*	*4A*	*2x*		*M*
▪ *Ticket Header*	*4B*	*0..1024a*		*O*
▪ *Ticket Body*	*4C*	*0..2048a*		*M*
▪ *Ticket Footer*	*4D*	*0..1024a*		*O*
• *Result Code*	*16*	*4x*		*M*
• Transparent Mode	3F01	GRP	O	O
○ Transparent Mode	1F01	2x	M	
○ Card UID	1F10	2..80x		O
○ Card ATR	1F11	2..80x		O
○ Card Type	1F12	2x		O

Table 1: Transaction request and response message structure

2.1.1 Transparent Mode Group

The transparent mode group contains the data required for transparent mode operation. In the request the “transparent mode” field indicates that the terminal should also detect custom (non-EMV) cards.

In the response, the “Transparent Mode” group will be only returned when a custom (non-EMV) card was detected. The “Transparent Mode” group contains information about the presented card and transparent mode is started. Now the ECR can send “transparent modes” messages to communicate with the card. The transaction result field will indicate “transparent mode” and no tickets are returned.

When an EMV card is presented, no “transparent mode” group is returned and the terminal will perform a regular transaction returning the transaction result and tickets.

2.1.1.1 Transparent Mode

When 1 the transparent mode field indicates that the terminal should start detecting custom cards for transparent mode. When 0, the terminal will only detect regular EMV cards (bank cards) and would be the same as not sending the “Transparent Mode” group at all.

2.1.1.2 Card UID

This is the UID of the presented custom card.

2.1.1.3 Card ATR

This is the ATR of the presented custom card, if available.

2.1.1.4 Card Type

This is the type of the presented card. This is the card type (SAK) value according to [2]. See also [3] for relevant AQA and SAK values.

2.1.2 Transaction Result

This is the result of the transaction. The possible values are given in Table 2. The *italic* rows are the existing codes. The non-italic rows are the new “transparent mode” codes.

Name	ID	Description
Approved	*0*	*The transaction succeeded.*
Error	*1*	*The transaction failed. See the Result Code for reason.*
TransparentMode	5	A non-EMV card was presented and transparent mode was started.

Table 2: Transaction result codes (transaction response)

2.2 Transparent Mode Request and Response

Instead of starting a regular transaction where both EMV (bank) cards and custom cards are detected, the “Transparent Mode Request” starts just detecting custom cards. When a custom card is detected the terminal enters transparent mode. Presenting EMV cards will result in an error.

Table 3 describes the structure of the *TransparentModeRequest* and *TransparentModeResponse* messages.

			Transparent Mode	
Data Element	Tag	Format	C-	R-
Message Body	E2	GRP	M	M
• Transparent Mode	1F01	2x	M	M
• Card Detection Time-out	1F02	4x	O	
• Card UID	1F10	2..80x		O
• Card ATR	1F11	2..80x		O
• Card Type	1F12	2x		O
• UI Control	3F02	GRP	O	
○ Display Text	1F20	0..80a	O	
○ Beep Type	1F21	2x	O	
○ LED Control	1F22	8b	O	
• Result Code	16	4x		M

Table 3: Intermediate result request and response message structure

2.2.1 Transparent Mode

In the request it indicates to start detecting custom cards and entering transparent mode (1) or when already in transparent mode, to stop the transparent mode (0).
In the response it indicates if the terminal is currently in transparent mode (1) or not (0).

2.2.2 Card Detection Time-out

This is the card detection time-out in seconds when asked to detect a custom card and enter transparent mode. When no card is presented within the time-out period, a “time-out” error is returned.
When no time-out is specified the default time-out will be used (5 seconds).

2.2.3 Card UID

This is the UID of the presented custom card.

2.2.4 Card ATR

This is the ATR of the presented custom card, if available.

2.2.5 Card Type

This is the type of the presented card. This is the card type SAK value according to [2]. See also [3] for relevant AQA and SAK values.

2.2.6 UI Control (Group)

This group controls the UI elements of the terminal (display, LEDs and buzzer).

2.2.6.1 Display Text

This is the text to be displayed on the terminal. The possible number of characters per line and the number of lines is dependent on the terminal used. For example on an unattended terminal (iUC180, iUN250) there are 4 lines with 16 characters per line while an iCT250 desktop terminal has more lines and uses proportional fonts where the number of characters depends on the characters used.

Also what happens when the text is too long for a line depends on the terminal type. On an unattended terminal, the line continues at the next line. On an iCT250 the lines are centered and cut-off at both sides. Similar when the number of lines is too big. On the unattended terminals the extra lines are not displayed, on an iCT250 the text is centered vertically and will be cut-off at both the top and bottom.

To send multiple lines, the lines must be separated by a '\n' (10) control character. Windows line endings "\r\n" must be replaced by only a '\n'.

2.2.6.2 Beep Type

The beep type determines which beep to sound. The possible beep type values are described in Table 4.

Name	ID	Description
Success	0	Three short success beeps.
Failure	1	One long failure beep.
Attention	2	One short attention beep.

Table 4: Beep types.

2.2.6.3 LED Control

The LED control determines which LEDs to turn on/off. This is a bit field thus multiple LEDs can be specified. When a bit is set, the corresponding LED is turned on; when a bit is not set the corresponding LED is turned off. Table 5 describes the possible values.

Bit	Meaning
1	LED 1 (left most LED).
2	LED 2.
3	LED 3.
4	LED 4 (right most LED).
5-8	- Reserved for future use – (currently always 0).

Table 5: LED control bit meanings

2.2.7 Result Code

This is the result of the request. See Appendix A.2 (Table 10) for the possible “transparent mode” result codes.

2.3 UI Control Request and Response

The *TmUiControlRequest* message is to control the UI elements of the terminal (display, LEDs and buzzer) while in transparent mode (after a card is detected).

Table 6 describes the structure of the *TmUiControlRequest* and *TmUiControlResponse* messages.

			TM UI Control	
Data Element	Tag	Format	C-	R-
Message Body	E2	GRP	M	M
• Display Text	1F20	0..80a	O	
• Beep Type	1F21	2x	O	
• LED Control	1F22	8b	O	
• Result Code	16	4x		M

Table 6: UI control request and response message structure

2.3.1 Display Text

Same as described in section 2.2.6.1.

2.3.2 Beep Type

Same as described in section 2.2.6.2.

2.3.3 LED Control

Same as described in section 2.2.6.3.

2.3.4 Result Code

This is the result of the request. See Appendix A.2 (Table 10) for the possible “transparent mode” result codes.

2.4 Authenticate Request and Response

The *TmAuthenticateRequest* is sent in transparent mode to authorise against the card. This is needed before data can be read from the card or written to the card.

Table 7 describes the structure of the *TmAuthenticateRequest* and *TmAuthenticateResponse* messages.

			TM Authenticate	
Data Element	Tag	Format	C-	R-
Message Body	E2	GRP	M	M
• Key	1F30	0..32x	M	
• Result Code	16	4x		M

Table 7: Authenticate request and response message structure

2.4.1 Key

This is the security key to use for authenticating to the card. The key is usually encrypted (contact Chess pT for the details).

2.4.2 Result Code

This is the result of the request. See Appendix A.2 (Table 10) for the possible “transparent mode” result codes.

2.5 Read Card Data Request and Response

The *TmReadCardDataRequest* is sent in transparent mode to read data from the card (after authentication).
Card data is usually accessed in pages but the interface makes it look like a random access memory block and translation to pages is done by the terminal.

Table 8 describes the structure of the *TmReadCardDataRequest* and *TmReadCardDataResponse* messages.

			TM Read Card Data	
Data Element	Tag	Format	C-	R-
Message Body	E2	GRP	M	M
• Address	1F31	4x	M	
• Data Size	1F32	4x	M	
• Data	1F33	0..131072x		O
• Result Code	16	4x		M

Table 8: Read Card Data request and response message structure

2.5.1 Address

This is the card “memory” address to read data from.

2.5.2 Data Size

In the request, this is the number of bytes to read.

2.5.3 Data

This is the data read from the card.

2.5.4 Result Code

This is the result of the request. See Appendix A.2 (Table 10) for the possible “transparent mode” result codes.

A Enumerations

A.1 Message Types

The message type indicates what message is stored in the body of a message. Table 9 describes the messages used in “transparent mode” with their code, the direction (from terminal to ECR or from ECR to terminal) and from which version (Protocol ID) the message is implemented.

Name	ID (dec)	Description	Source	Dest	Version
Transaction Request	48	Request to perform a transaction.	ECR	Terminal	1
Transaction Response	49	Result of the transaction.	Terminal	ECR	1
Transparent Mode Request	1000	Request to start or stop transparent mode.	ECR	Terminal	2
Transparent Mode Response	1001	Result of the request to start/stop transparent mode.	Terminal	ECR	2
TM UI Control Request	1002	Request to update the user interface (buzzer, display, LEDs) when in transparent mode.	ECR	Terminal	2
TM UI Control Response	1003	Result of the UI request.	Terminal	ECR	2
TM Authenticate Request	1004	Request to authenticate to the card when in transparent mode.	ECR	Terminal	2
TM Authenticate Response	1005	Result of the authenticate request.	Terminal	ECR	2
TM Read Card Data Request	1006	Request to read data from the card when in transparent mode.	ECR	Terminal	2
TM Read Card Data Response	1007	Result of the read card data request.	Terminal	ECR	2
TM Write Card Data Request	1008	Request to write data to the card when in transparent mode.	ECR	Terminal	2
TM Write Card Response	1009	Result of the write card data request.	Terminal	ECR	2

Table 9: Transparent mode message types

A.2 Result Codes

There are additional error-codes defined for transparent mode related errors which are described in. Table 10.

Name	ID (decimal)	Description
ResultOK	0	All OK, no problems.
TmError	1000	Unspecified transparent mode error.
TmNotAvailable	1001	The transparent mode is not available/enabled on the terminal.
TmTimeout	1002	A time-out occurred.
TmNotInTransparentMode	1003	A transparent mode message was sent while the terminal was not in transparent mode.
TmAuthenticationFailed	1004	Failed to authenticate with the card.
TmNotAuthenticated	1005	Request failed because authentication was not done.
TmReadDataFailed	1006	Failed to read data from the card.
TmWriteDataFailed	1007	Failed to write data to the card.
TmAlreadyInTransparentMode	1008	The terminal was already in transparent mode.

Table 10: Result codes